

Business Problem Statement

Project Title

AI-Powered Credit Risk Analytics & Loan Default Prediction Platform

Overview

Financial institutions generate revenue by issuing loans, but every approved loan carries the risk of borrower default. Loan defaults can lead to financial losses, increased recovery costs, and reduced profitability. At the same time, rejecting creditworthy applicants may result in lost business opportunities and reduced customer growth.

Home Credit serves customers with limited or non-traditional credit histories, making credit risk assessment more challenging. To balance financial inclusion with risk management, the company requires a data-driven approach to evaluate borrower risk and support lending decisions.

Business Problem

Home Credit has access to large volumes of customer, credit bureau, loan application, and repayment data. However, converting this data into actionable insights for identifying high-risk borrowers remains a significant challenge.

The core business question is:

How can historical customer and credit data be leveraged to predict loan default risk, improve lending decisions, and reduce financial losses?

Without an effective risk assessment framework, the organization may experience higher default rates, poor portfolio performance, and inefficient lending decisions.

Project Objectives

The project aims to:

- Identify customers with a high probability of default.
- Develop predictive models to estimate default risk.
- Segment customers into risk categories.
- Support data-driven lending decisions.
- Improve portfolio quality and reduce credit losses.
- Generate explainable AI-powered risk insights.

Key Business Questions

- Which customer characteristics contribute most to loan defaults?
- How does credit history influence repayment behavior?
- Which customer segments represent the highest risk?

- What factors are the strongest predictors of default?
- How can lending decisions be optimized using analytics?

Proposed Solution

To address these challenges, the project will develop an end-to-end Credit Risk Analytics Platform incorporating:

- Descriptive Analytics for understanding customer and loan behavior.
- Diagnostic Analytics for identifying drivers of default.
- Predictive Analytics using machine learning models.
- Power BI Dashboards for portfolio monitoring and risk analysis.
- AI-Powered Risk Assistant for explainable lending recommendations.

Expected Outcomes

The project is expected to:

- Predict loan default probabilities.
- Identify high-risk borrowers before approval.
- Improve lending decision accuracy.
- Reduce financial losses.
- Enhance portfolio performance.
- Support transparent and data-driven credit risk management.

Business Value

By combining analytics, machine learning, business intelligence, and AI, the solution will help Home Credit improve risk management practices while maintaining sustainable business growth and responsible lending.